

Mathematical Analysis Notes: Tutorial 1 and Later

Tutorial 1

1 Proof, logic, and limit language

1.1 Basic terminology

- An axiom is a statement, rule, or principle that is accepted as true without needing proof.
- A theorem is an important statement established by proof; a less central result is often called a proposition or lemma.
- A proof is a finite, logically valid argument using definitions, axioms, and results already proved.
- A statement of the form $P \Rightarrow Q$ is a conditional statement: P is the hypothesis and Q is the conclusion.

1.2 The ε - δ definition

Here δ controls how close the input x is to x_0 , while ε prescribes how close $f(x)$ must be to ℓ .

A standard proof has the following shape:

1. Let $\varepsilon > 0$ be arbitrary.
2. Choose an explicit $\delta > 0$, usually by first estimating $|f(x) - \ell|$ in terms of $|x - x_0|$.
3. Assume $0 < |x - x_0| < \delta$ and verify $|f(x) - \ell| < \varepsilon$.

1.3 Direct proof, contradiction, and contraposition

Direct proof To prove a statement of the form

$$P \Rightarrow Q,$$

assume that P is true and use definitions, axioms, and previously proved results until Q follows. This is the default method for most ε - δ arguments.

Proof by contradiction To prove a statement S :

1. Assume its negation, $\neg S$.
2. Derive a contradiction, such as $R \wedge \neg R$, an impossible inequality, or a conflict with a known theorem.
3. Conclude that $\neg S$ is false, so S is true.

For a uniqueness statement, assume that two distinct objects both satisfy the defining property. For a nonexistence statement, assume that the object exists and use its required properties to obtain a contradiction.

Example: irrationality of $\sqrt{2}$ Suppose, for contradiction, that

$$\sqrt{2} = \frac{p}{q},$$

where p, q are positive integers with no common factor. Squaring gives

$$p^2 = 2q^2.$$

Thus p^2 is even, so p is even. Write $p = 2r$. Substitution gives

$$4r^2 = 2q^2 \implies q^2 = 2r^2.$$

Therefore q is also even. Hence p and q have the common factor 2, contradicting the assumption that p/q is in lowest terms. Consequently,

$$\sqrt{2} \notin \mathbb{Q}.$$

Proof by contraposition To prove

$$P \Rightarrow Q,$$

it is enough to prove the logically equivalent statement

$$\neg Q \Rightarrow \neg P.$$

This method is especially useful when the negation of the conclusion gives more concrete information than the conclusion itself.

1.4 Mathematical induction

Let $P(n)$ be a statement for $n \in \mathbb{Z}_{>0}$. The principle of mathematical induction says that if:

1. **Base case:** $P(1)$ is true;
2. **Inductive step:** for every positive integer n , assuming $P(n)$ is true implies that $P(n+1)$ is true;

then $P(n)$ is true for every positive integer n .

In the inductive step, clearly state the **induction hypothesis** $P(n)$ before proving $P(n+1)$. Do not assume the statement that is being proved.

Example: sum of the first n positive integers We prove that

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \quad (n \geq 1).$$

For $n = 1$, both sides equal 1. Now assume the formula holds for $n = k$:

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}.$$

Then

$$\begin{aligned} 1 + 2 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

Thus, the statement for k implies the statement for $k+1$, so the formula holds for every $n \geq 1$.

Strong induction The principle of strong induction replaces the induction hypothesis $P(n)$ by

$$P(1), P(2), \dots, P(n) \implies P(n+1).$$

It is useful when the case $n+1$ depends on several earlier cases rather than only on the immediately preceding case.

Ordinary induction, strong induction, and the well-ordering principle are equivalent principles. The well-ordering principle states that every nonempty subset of $\mathbb{Z}_{>0}$ has a least element.

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Exercise 1.1.26

Any polynomial of odd degree with real coefficients has at least one real root.

Proof

Let

$$P(x) = \sum_{k=0}^n a_k x^k$$

be a polynomial of degree n with real coefficients, where n is an odd integer and $a_n \neq 0$. Without loss of generality, assume that $a_n > 0$. If $a_n < 0$, we can apply the same argument to $-P$.

Let

$$C = \max_{0 \leq k \leq n-1} |a_k|.$$

For $|x| \geq 1$, we have

$$\begin{aligned} |P(x) - a_n x^n| &= \left| \sum_{k=0}^{n-1} a_k x^k \right| \\ &\leq C \sum_{k=0}^{n-1} |x|^k \\ &\leq Cn|x|^{n-1} = \frac{Cn}{|a_n||x|} |a_n x^n|. \end{aligned}$$

Choose

$$R > \max \left\{ 1, \frac{2Cn}{|a_n|} \right\}.$$

Then, whenever $|x| \geq R$,

$$|P(x) - a_n x^n| < \frac{1}{2} |a_n x^n|.$$

Since n is odd, $R^n > 0$ and $(-R)^n < 0$. For $x = R$, we obtain

$$\begin{aligned} P(R) &\geq a_n R^n - |P(R) - a_n R^n| \\ &> a_n R^n - \frac{1}{2} |a_n R^n| = \frac{1}{2} a_n R^n > 0. \end{aligned}$$

For $x = -R$, we obtain

$$\begin{aligned} P(-R) &\leq a_n (-R)^n + |P(-R) - a_n (-R)^n| \\ &< a_n (-R)^n + \frac{1}{2} |a_n (-R)^n| = \frac{1}{2} a_n (-R)^n < 0. \end{aligned}$$

Thus,

$$P(-R) < 0 < P(R).$$

Because P is continuous on $[-R, R]$, the Intermediate Value Theorem gives some $c \in [-R, R]$ such that

$$P(c) = 0.$$

Therefore, P has at least one real root. $\square$

Proof of the Intermediate Value Theorem

Let f be continuous on $[a, b]$, and suppose first that

$$f(a) < y_0 < f(b).$$

Define

$$S = \{x \in [a, b] : f(x) \leq y_0\}.$$

The set S is nonempty because $a \in S$, and it is bounded above by b . By the completeness of $\mathbb{R}$, the supremum

$$c = \sup S$$

exists. Since $f(a) < y_0 < f(b)$, we have $a < c < b$.

We claim that $f(c) = y_0$. Suppose first that $f(c) < y_0$. By continuity of f at c , there exists $\delta > 0$ such that

$$|x - c| < \delta \implies f(x) < y_0.$$

Choosing $x > c$ sufficiently close to c gives $x \in S$, contradicting the fact that c is an upper bound of S .

On the other hand, suppose that $f(c) > y_0$. By continuity of f at c , there exists $\delta > 0$ such that

$$|x - c| < \delta \implies f(x) > y_0.$$

Since $c = \sup S$, there is some $x \in S$ with $c - \delta < x \leq c$. This contradicts $f(x) > y_0$.

Therefore neither $f(c) < y_0$ nor $f(c) > y_0$ is possible, and hence

$$f(c) = y_0.$$

If $y_0 = f(a)$ or $y_0 = f(b)$, the conclusion is immediate by taking $c = a$ or $c = b$. If $f(b) < f(a)$, apply the same argument to $-f$. Therefore, for every value y_0 between $f(a)$ and $f(b)$, there exists $c \in [a, b]$ such that

$$f(c) = y_0.$$

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Limits Involving Infinity

Finite limits as $x \rightarrow +\infty$

We say that

$$\lim_{x \rightarrow +\infty} f(x) = L$$

if, for every $\varepsilon > 0$, there exists $M > 0$ such that

$$x > M \implies |f(x) - L| < \varepsilon.$$

In words, $f(x)$ becomes arbitrarily close to L when x is sufficiently large.

Similarly,

$$\lim_{x \rightarrow -\infty} f(x) = L$$

if, for every $\varepsilon > 0$, there exists $M < 0$ such that

$$x < M \implies |f(x) - L| < \varepsilon.$$

Examples

For the reciprocal function,

$$\lim_{x \rightarrow +\infty} \frac{1}{x} = 0, \quad \lim_{x \rightarrow -\infty} \frac{1}{x} = 0.$$

Indeed, if $|x| > 1/\varepsilon$, then

$$\left| \frac{1}{x} - 0 \right| = \frac{1}{|x|} < \varepsilon.$$

For a rational function whose numerator has lower degree than its denominator,

$$\lim_{x \rightarrow \pm\infty} \frac{P(x)}{Q(x)} = 0, \quad \deg P < \deg Q.$$

Infinite limits as $x \rightarrow c$

We write

$$\lim_{x \rightarrow c} f(x) = +\infty$$

if, for every $A > 0$, there exists $\delta > 0$ such that

$$0 < |x - c| < \delta \implies f(x) > A.$$

Similarly,

$$\lim_{x \rightarrow c} f(x) = -\infty$$

if, for every $B > 0$, there exists $\delta > 0$ such that

$$0 < |x - c| < \delta \implies f(x) < -B.$$

For example,

$$\lim_{x \rightarrow 0} \frac{1}{x^2} = +\infty,$$

because $1/x^2$ can be made larger than any prescribed positive number by choosing x sufficiently close to 0, with $x \neq 0$.

Infinite limits as $x \rightarrow \pm\infty$

The notation

$$\lim_{x \rightarrow +\infty} f(x) = +\infty$$

means that for every $A > 0$, there exists $M > 0$ such that

$$x > M \implies f(x) > A.$$

The other cases are defined analogously. For instance, if n is odd and $a_n > 0$, then

$$\lim_{x \rightarrow +\infty} a_n x^n = +\infty, \quad \lim_{x \rightarrow -\infty} a_n x^n = -\infty.$$

Exercise 1.1.30

(a)

Prove, using the N - δ language, that

$$\lim_{x \rightarrow 0^+} \frac{1}{\sqrt[3]{x}} = +\infty.$$

(b)

Find the limit

$$\lim_{x \rightarrow +\infty} (\sqrt{x+1} - \sqrt{x-1}),$$

and prove your result using the ε - N language.

Solution

(a) Let $N > 0$ be arbitrary. Choose

$$\delta = \frac{1}{N^3}.$$

If $0 < x < \delta$, then

$$\sqrt[3]{x} < \sqrt[3]{\delta} = \frac{1}{N}.$$

Therefore,

$$\frac{1}{\sqrt[3]{x}} > \frac{1}{\sqrt[3]{\delta}} = N.$$

Hence, for every $N > 0$, there exists $\delta > 0$ such that

$$0 < x < \delta \implies \frac{1}{\sqrt[3]{x}} > N.$$

Thus,

$$\boxed{\lim_{x \rightarrow 0^+} \frac{1}{\sqrt[3]{x}} = +\infty}.$$

(b) Let

$$f(x) = \sqrt{x+1} - \sqrt{x-1}.$$

For $x \geq 1$, rationalizing gives

$$f(x) = \frac{(x+1) - (x-1)}{\sqrt{x+1} + \sqrt{x-1}} = \frac{2}{\sqrt{x+1} + \sqrt{x-1}}.$$

Since $\sqrt{x+1} + \sqrt{x-1} \rightarrow +\infty$ as $x \rightarrow +\infty$, we obtain

$$\lim_{x \rightarrow +\infty} f(x) = 0.$$

To prove this using the ε - N language, let $\varepsilon > 0$ be given. Choose

$$N = 1 + \frac{1}{\varepsilon^2}.$$

If $x > N$, then $x > 1$ and

$$0 < f(x) = \frac{2}{\sqrt{x+1} + \sqrt{x-1}} \leq \frac{1}{\sqrt{x-1}}.$$

Moreover, $x > N$ implies $x - 1 > 1/\varepsilon^2$, so

$$|f(x) - 0| = f(x) \leq \frac{1}{\sqrt{x-1}} < \varepsilon.$$

Therefore, for every $\varepsilon > 0$, there exists $N > 0$ such that

$$x > N \quad \Rightarrow \quad \left| \sqrt{x+1} - \sqrt{x-1} \right| < \varepsilon.$$

Hence,

$$\boxed{\lim_{x \rightarrow +\infty} (\sqrt{x+1} - \sqrt{x-1}) = 0}.$$

Exercise 1.1.28

All the Limit Laws remain true when we replace

$$\lim_{x \rightarrow c}$$

by

$$\lim_{x \rightarrow +\infty} \quad \text{or} \quad \lim_{x \rightarrow -\infty}.$$

For example, if

$$\lim_{x \rightarrow +\infty} f(x) = L, \quad \lim_{x \rightarrow +\infty} g(x) = M,$$

then

$$\lim_{x \rightarrow +\infty} (f(x) + g(x)) = L + M,$$

and, provided $M \neq 0$,

$$\lim_{x \rightarrow +\infty} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

The same statements hold as $x \rightarrow -\infty$.

Asymptotes

The graph of $y = f(x)$ on an interval I is

$$\{(x, y) : x \in I, y = f(x)\}.$$

A line is called an **asymptote** of the graph if the distance between the graph and the line approaches 0 as the point on the graph moves to infinity or approaches a finite endpoint.

Horizontal asymptotes

A line $y = b$ is a horizontal asymptote of $y = f(x)$ as $x \rightarrow +\infty$ if

$$\lim_{x \rightarrow +\infty} f(x) = b.$$

Similarly, $y = b$ is a horizontal asymptote as $x \rightarrow -\infty$ if

$$\lim_{x \rightarrow -\infty} f(x) = b.$$

Example 1.1.32 Find the horizontal asymptotes of

$$f(x) = \frac{\sin x + \cos x}{1 + x^2},$$

and

$$g(x) = \sqrt{1 - \frac{1}{x^2}}, \quad |x| \geq 1.$$

Since $|\sin x + \cos x| \leq 2$,

$$\left| \frac{\sin x + \cos x}{1 + x^2} \right| \leq \frac{2}{1 + x^2} \rightarrow 0.$$

Thus, $y = 0$ is a horizontal asymptote of f as $x \rightarrow \pm\infty$.

For g ,

$$\lim_{x \rightarrow \pm\infty} g(x) = \sqrt{1 - \lim_{x \rightarrow \pm\infty} \frac{1}{x^2}} = 1.$$

Thus, $y = 1$ is a horizontal asymptote of g as $x \rightarrow \pm\infty$.

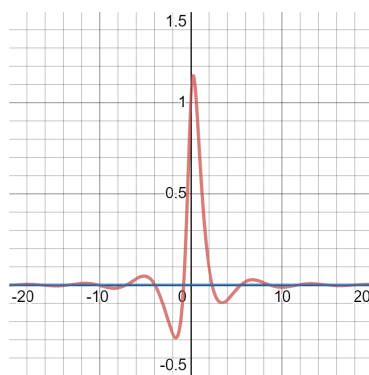


Figure: $f = \frac{\sin X + \cos X}{1 + X^2}$

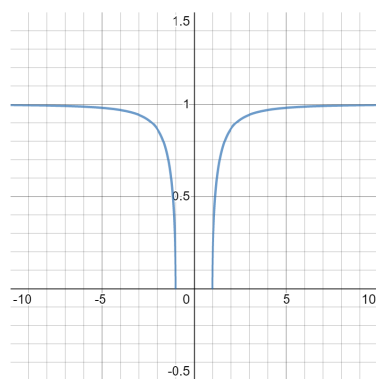


Figure: $g = \sqrt{1 - \frac{1}{x^2}}$

Sketches of horizontal asymptotes

Vertical asymptotes

A line $x = a$ is a vertical asymptote of $y = f(x)$ as $x \rightarrow a^+$ if

$$\lim_{x \rightarrow a^+} f(x) = +\infty \quad \text{or} \quad \lim_{x \rightarrow a^+} f(x) = -\infty.$$

The analogous definition applies as $x \rightarrow a^-$. In the notation $\pm\infty$, the sign may be either $+$ or $-$.

Example 1.1.33 Consider

$$h(x) = \sqrt{1 - \frac{\sin x}{x^2}},$$

on its domain

$$(-\infty, 0) \cup [a, +\infty),$$

where the boundary point $a > 0$ is determined by

$$a^2 = \sin a,$$

where $0 < a < 1$. As $x \rightarrow 0^-$, we have $\sin x \sim x$, and therefore

$$1 - \frac{\sin x}{x^2} \sim 1 - \frac{1}{x} \rightarrow +\infty.$$

Consequently,

$$\lim_{x \rightarrow 0^-} h(x) = +\infty,$$

so $x = 0$ is a vertical asymptote. Near $x = a$ from the right, the expression under the square root approaches 0, so $x = a$ is not a vertical asymptote.

This example emphasizes that a domain endpoint is not automatically a vertical asymptote: the function itself must tend to $\pm\infty$.

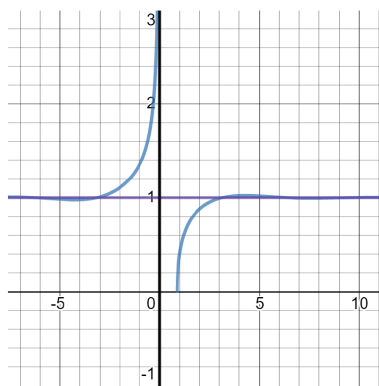


Figure: $f = \sqrt{1 - \frac{\sin x}{x^2}}$

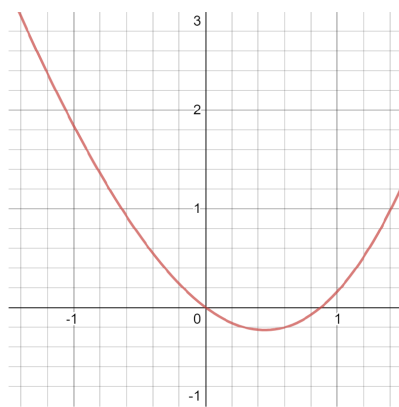


Figure: $y = x^2 - \sin x$

Sketches of vertical asymptotes

Oblique asymptotes

A line

$$y = ax + b, \quad a \neq 0,$$

is an oblique asymptote of $y = f(x)$ as $x \rightarrow +\infty$ if

$$\lim_{x \rightarrow +\infty} [f(x) - ax - b] = 0.$$

The definition as $x \rightarrow -\infty$ is analogous.

When the limits exist, the coefficients can be found from

$$a = \lim_{x \rightarrow \pm\infty} \frac{f(x)}{x}, \quad b = \lim_{x \rightarrow \pm\infty} [f(x) - ax].$$

Example 1.1.34 Find the oblique asymptote of

$$f(x) = \frac{x^2 - 3}{2x - 4}.$$

Polynomial division gives

$$\begin{aligned} f(x) &= \frac{x^2 - 4 + 1}{2(x - 2)} \\ &= \frac{1}{2}(x + 2) + \frac{1}{2(x - 2)}. \end{aligned}$$

Therefore,

$$f(x) - \left(\frac{x}{2} + 1\right) = \frac{1}{2(x - 2)} \rightarrow 0 \quad (x \rightarrow \pm\infty).$$

The oblique asymptote is

$$y = \frac{x}{2} + 1.$$

Also, $x = 2$ is a vertical asymptote.

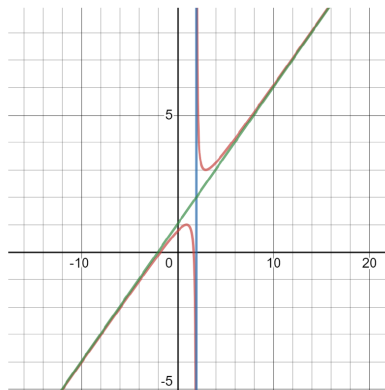


Figure: $f = \frac{x^2-3}{2x-4}$

Example 1.1.35 Let

$$f(x) = \frac{x^3}{(x+2)(x-1)}.$$

The vertical asymptotes are

$$x = -2, \quad x = 1.$$

For the oblique asymptote,

$$\frac{f(x)}{x} = \frac{x^2}{(x+2)(x-1)} \rightarrow 1,$$

and

$$f(x) - x = \frac{-x^2 + 2x}{(x+2)(x-1)} \rightarrow -1.$$

Hence the oblique asymptote is

$$y = x - 1$$

as $x \rightarrow \pm\infty$.

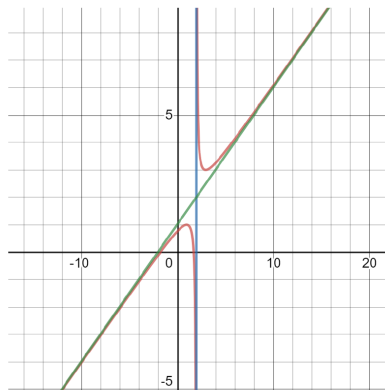


Figure: $f = \frac{x^2-3}{2x-4}$

Exercise 1.1.36

Let

$$f(x) = \frac{1}{x} \sin \frac{1}{x}.$$

Does the graph of f have a vertical asymptote at $x = 0$? Explain your answer using the definition of a vertical asymptote.

Since

$$\left| \frac{1}{x} \sin \frac{1}{x} \right| \leq \frac{1}{|x|},$$

this estimate alone does not determine an infinite limit. To see the oscillation explicitly, define

$$x_n = \frac{1}{\frac{\pi}{2} + 2\pi n}, \quad y_n = \frac{1}{\frac{3\pi}{2} + 2\pi n}.$$

Then $x_n, y_n \rightarrow 0^+$ and

$$f(x_n) = \frac{\pi}{2} + 2\pi n \rightarrow +\infty,$$

while

$$f(y_n) = -\left(\frac{3\pi}{2} + 2\pi n\right) \rightarrow -\infty.$$

Thus, as $x \rightarrow 0^+$, the function tends neither to $+\infty$ nor to $-\infty$. Applying the same argument to $-x_n$ and $-y_n$ gives the same conclusion as $x \rightarrow 0^-$. Therefore, $x = 0$ is not a vertical asymptote in the usual one-sided sense.

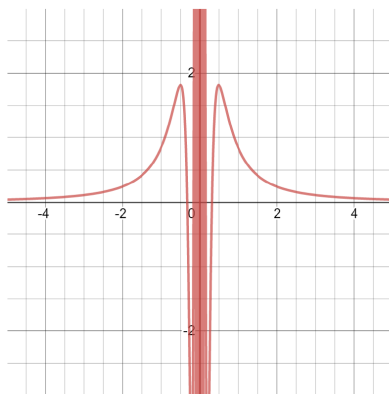


Figure: $f = \frac{1}{x} \sin \frac{1}{x}$

More Exercises

Find the limits and use the ε - N language to prove your results.

(a)

$$\lim_{x \rightarrow +\infty} \sqrt{x} (\sqrt{x+1} - \sqrt{x-1}).$$

(b)

$$\lim_{x \rightarrow +\infty} \sqrt{x + \sqrt{x + \frac{\sqrt{x}}{\sqrt{x+1}}}}.$$

Lecture 7

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1.7 Sequences and limits

Definition 1.1.12 (Limit of a sequence)

A sequence $\{a_n\}$ converges to L if for every $\varepsilon > 0$ there exists an integer N such that

$$n > N \implies |a_n - L| < \varepsilon.$$

If no such number L exists, we say that the sequence diverges.

We write

$$\lim_{n \rightarrow \infty} a_n = L, \quad \text{or} \quad a_n \rightarrow L.$$

Exercise 1.1.39

Show that if a sequence converges, then the limit is unique.

Proof Suppose that

$$a_n \rightarrow L_1, \quad a_n \rightarrow L_2.$$

Let $\varepsilon > 0$ be arbitrary. Then there exist N_1, N_2 such that

$$n > N_1 \Rightarrow |a_n - L_1| < \varepsilon/2,$$

and

$$n > N_2 \Rightarrow |a_n - L_2| < \varepsilon/2.$$

For $n > \max\{N_1, N_2\}$,

$$|L_1 - L_2| \leq |a_n - L_1| + |a_n - L_2| < \varepsilon.$$

Since this holds for every $\varepsilon > 0$, we must have $L_1 = L_2$.

Hence the limit is unique.

Example 1.1.40

Investigate the convergence or divergence of the following sequences.

(a) $(-1)^n$ This sequence alternates between -1 and 1 , so it does not approach any single number. Therefore,

$$(-1)^n \text{ diverges.}$$

(b) $\frac{\sin n - n^2}{n^3 + 1}$ Since

$$\frac{\sin n - n^2}{n^3 + 1} = \frac{\sin n}{n^3 + 1} - \frac{n^2}{n^3 + 1},$$

and both terms go to 0, we have

$$\lim_{n \rightarrow \infty} \frac{\sin n - n^2}{n^3 + 1} = 0.$$

(c) $\frac{(-1)^{[\sqrt{n}]}}{n}$ Here $[\sqrt{n}]$ is the integer part of $\sqrt{n}$, and the numerator is always ± 1 . Since dividing by n makes the magnitude go to 0,

$$\lim_{n \rightarrow \infty} \frac{(-1)^{[\sqrt{n}]}}{n} = 0.$$

Bounded sequences

Definition

A sequence $\{a_n\}$ is called:

- bounded above if there exists a constant M such that $a_n \leq M$ for all n ;
- bounded below if there exists a constant m such that $a_n \geq m$ for all n ;
- bounded if there exists a constant $C > 0$ such that $|a_n| \leq C$ for all n .

Equivalently, $\{a_n\}$ is bounded iff there exist m, M with

$$m \leq a_n \leq M \quad \forall n \geq 1.$$

Theorem

Every convergent sequence is bounded.

Proof If $a_n \rightarrow L$, then for $\varepsilon = 1$ there exists N such that

$$n > N \Rightarrow |a_n - L| < 1.$$

Thus for all $n > N$,

$$|a_n| \leq |L| + 1.$$

Let

$$C = \max\{|a_1|, \dots, |a_N|, |L| + 1\}.$$

Then $|a_n| \leq C$ for all n , so the sequence is bounded.

Divergence to infinity

Definition

We say that $a_n \rightarrow +\infty$ if for every real number M there exists an integer N such that

$$n \geq N \Rightarrow a_n > M.$$

We say that $a_n \rightarrow -\infty$ if for every real number $m > 0$ there exists an integer N such that

$$n \geq N \Rightarrow a_n < -m.$$

Example 1.1.41

Investigate the behavior of the following sequences as $n \rightarrow \infty$.

(a) $(-1)^n n$ This sequence alternates sign and grows in magnitude, so it is unbounded and does not converge to a finite limit. It also does not tend to $+\infty$ or $-\infty$ in the usual sense because the signs alternate.

(b) $a^n/n!$ **with** $a > 0$ For fixed $a > 0$, the factorial grows faster than any exponential, so

$$\frac{a^n}{n!} \rightarrow 0.$$

(c) $n^n/n!$ Since

$$\frac{n^n}{n!} = \prod_{k=1}^n \frac{n}{k},$$

this sequence grows very rapidly, so

$$\frac{n^n}{n!} \rightarrow +\infty.$$

Limit laws for sequences

If

$$\lim_{n \rightarrow \infty} a_n = A, \quad \lim_{n \rightarrow \infty} b_n = B,$$

then the following hold:

1. $\lim_{n \rightarrow \infty} (a_n + b_n) = A + B$;
2. $\lim_{n \rightarrow \infty} (a_n - b_n) = A - B$;
3. $\lim_{n \rightarrow \infty} (ka_n) = kA$ for any constant k ;
4. $\lim_{n \rightarrow \infty} (a_n b_n) = AB$;
5. if $B \neq 0$, then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{A}{B}.$$

Exercise 1.1.43

Let $\{a_n\}$ and $\{b_n\}$ be convergent sequences and assume $a_n \geq b_n$ for all sufficiently large n . Prove that

$$\lim_{n \rightarrow \infty} a_n \geq \lim_{n \rightarrow \infty} b_n.$$

Proof

Let

$$A = \lim_{n \rightarrow \infty} a_n, \quad B = \lim_{n \rightarrow \infty} b_n.$$

Suppose, for contradiction, that $A < B$. Set

$$\varepsilon = \frac{B - A}{2} > 0.$$

Then there exists N such that for all $n > N$,

$$|a_n - A| < \varepsilon, \quad |b_n - B| < \varepsilon.$$

Hence for $n > N$,

$$a_n < A + \varepsilon = \frac{A + B}{2},$$

and

$$b_n > B - \varepsilon = \frac{A + B}{2}.$$

So $a_n < b_n$ for all sufficiently large n , contradicting the assumption that $a_n \geq b_n$ eventually. Therefore,

$$A \geq B.$$

Exercise 1.1.43 (continued)

Assume $a_n \leq b_n \leq c_n$ for all $n \geq N_0$ and

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L.$$

Then

$$\lim_{n \rightarrow \infty} b_n = L.$$

Proof

Let $\varepsilon > 0$ be given. Since $a_n \rightarrow L$ and $c_n \rightarrow L$, there exist N_1, N_2 such that

$$n > N_1 \Rightarrow |a_n - L| < \varepsilon,$$

and

$$n > N_2 \Rightarrow |c_n - L| < \varepsilon.$$

Let $N = \max\{N_0, N_1, N_2\}$. Then for all $n > N$,

$$L - \varepsilon < a_n \leq b_n \leq c_n < L + \varepsilon.$$

Hence

$$|b_n - L| < \varepsilon,$$

for all $n > N$. Therefore, $b_n \rightarrow L$.

This is the sequence version of the Sandwich Theorem.

Exercise 1.1.44

Let $\{a_n\}$ be a sequence with $a_n \rightarrow L$, and let f be a function continuous at L and defined at all a_n . Then

$$f(a_n) \rightarrow f(L).$$

Proof

Let $\varepsilon > 0$ be given. Since f is continuous at L , there exists $\delta > 0$ such that

$$|x - L| < \delta \Rightarrow |f(x) - f(L)| < \varepsilon.$$

Because $a_n \rightarrow L$, there exists N such that

$$n > N \Rightarrow |a_n - L| < \delta.$$

Therefore, for all $n > N$,

$$|f(a_n) - f(L)| < \varepsilon.$$

So $f(a_n) \rightarrow f(L)$.

Proposition 1.1.13 (function limits and sequence limits)

Let I be an interval containing x_0 in its interior and let f be defined on $I \setminus \{x_0\}$. Then

$$\lim_{x \rightarrow x_0} f(x) = A$$

exists if and only if for every sequence $\{a_n\} \subset I \setminus \{x_0\}$ with $a_n \rightarrow x_0$, the limit

$$\lim_{n \rightarrow \infty} f(a_n)$$

exists and equals A .

Proof sketch

- If $f(x) \rightarrow A$ as $x \rightarrow x_0$, then for any sequence $a_n \rightarrow x_0$, the values $f(a_n)$ must also approach A .
- Conversely, if every sequence approaching x_0 produces the same limiting value A , then $f(x)$ cannot fail to approach A ; otherwise one could construct a sequence $x_n \rightarrow x_0$ with $|f(x_n) - A| \geq \varepsilon_0$ for all n , contradicting the assumption.

Thus, the limit of a function can be tested using sequences.

Exercise 1.1.45

Let f be a function defined on $[M, \infty)$. Prove:

(a)

$$\lim_{x \rightarrow \infty} f(x) = L$$

if and only if for every sequence $a_n \rightarrow \infty$,

$$\lim_{n \rightarrow \infty} f(a_n) = L.$$

(b)

$$\lim_{x \rightarrow \infty} f(x) = +\infty$$

if and only if for every sequence $a_n \rightarrow \infty$,

$$\lim_{n \rightarrow \infty} f(a_n) = +\infty.$$

These results show that the behavior of a function at infinity can be studied through sequences tending to infinity.

1.8 Least upper bound and greatest lower bound

Definition

Let $\{a_n\}$ be a sequence bounded above. A number L is called the least upper bound, or supremum, of $\{a_n\}$ if:

1. $a_n \leq L$ for all n ;
2. if M is any upper bound, then $L \leq M$.

We write

$$L = \sup_{n \geq 1} a_n.$$

Similarly, if $\{a_n\}$ is bounded below, a number l is called the greatest lower bound, or infimum, if:

1. $a_n \geq l$ for all n ;
2. if m is any lower bound, then $l \geq m$.

We write

$$l = \inf_{n \geq 1} a_n.$$

Remarks

- For the supremum: for every $\varepsilon > 0$, there exists n such that

$$a_n > L - \varepsilon.$$

- For the infimum: for every $\varepsilon > 0$, there exists n such that

$$a_n < l + \varepsilon.$$

Theorem 1.1.15

Any sequence bounded above has a supremum, and any sequence bounded below has an infimum.

This is the completeness property of the real numbers.

Example 1.1.47

Find the infimum and supremum of the following sequences.

(a) $\{(-1)^n/n\}_{n \geq 1}$

$$\inf = -\frac{1}{2}, \quad \sup = \frac{1}{2}.$$

(b) $\{(-1)^n n\}_{n \geq 1}$

This sequence is unbounded above and below, so

$$\sup = +\infty, \quad \inf = -\infty.$$

(c) $\{n(-1)^n\}_{n \geq 1}$

Again unbounded, so

$$\sup = +\infty, \quad \inf = -\infty.$$

(d) $a_1 = \sqrt{2}$, $a_{n+1} = \sqrt{2 + a_n}$

This sequence is increasing and bounded above by 2, so it converges to the positive solution of

$$x = \sqrt{2 + x},$$

namely

$$x^2 - x - 2 = 0 \Rightarrow x = 2.$$

Thus,

$$\sup = 2, \quad \inf = \sqrt{2}.$$

Lecture 8

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